BNN for Classification

General Principles

To model complex, non-linear boundaries between classes, we can extend the Bayesian Neural Network (BNN) architecture to classification tasks. While a regression BNN outputs a continuous value modeled with a Normal likelihood, a classification BNN outputs probabilities.

For binary classification, the final layer of the network condenses the hidden representations into a single output score. This score is then passed through a **sigmoid** activation (or used directly as the `logits` parameter) to map it to a probability between 0 and 1. We then use a **Bernoulli** or **Binomial** distribution to model the likelihood of the observed binary outcomes.

Considerations

> **Note**
>
> - **Uncertainty in Classification**: In standard neural networks, softmax or sigmoid outputs are often overconfident. By using a BNN, we obtain a posterior predictive distribution that reflects true uncertainty, leading to better-calibrated probabilities. Areas with little training data will appropriately show high uncertainty rather than confident misclassifications.
> - **Activation and Likelihood**: For binary classification, we pair the linear output of the network's final layer with a Binomial or Bernoulli likelihood. The final linear output acts as the **logit** input to the likelihood function, effectively applying a link function to constrain outputs to $[0, 1]$.
> - **Prior distributions**: Similar to regression BNNs, we apply weakly-informative priors (like `Normal(0, 1)`) to the weights of each layer.

Example

Below is an example code snippet demonstrating a *Bayesian Neural Network for classification* using the Bayesian Inference (**BI**) package. This example is inspired by stochastic variational inference tutorials and uses a synthetic nested moons dataset (typically generated via `make_moons`).

Python

```python
from BI import bi
import jax.numpy as jnp
from sklearn.datasets import make_moons

# Setup device------------------------------------------------
m = bi(platform='cpu')

# Generate Synthetic Data ------------------------------------
# Two interleaving half-moon shapes
X, Y = make_moons(n_samples=500, noise=0.25, random_state=42)

# Convert to JAX arrays
X = jnp.array(X)
Y = jnp.array(Y)

m.data_on_model = dict(X=X, Y=Y)

# Define model -----------------------------------------------
def model(X, Y, D_H1=4, D_H2=3):
    N, D_X = X.shape

    # First hidden layer: 2 input features -> 4 hidden units
    w1 = m.bnn.layer_linear(
        X,
        dist=m.dist.normal(0, 1, name='w1', shape=(D_X, D_H1)),
        activation='tanh'
    )

    # Second hidden layer: 4 hidden units -> 3 hidden units
    w2 = m.bnn.layer_linear(
        X=w1,
        dist=m.dist.normal(0, 1, name='w2', shape=(D_H1, D_H2)),
```

```
        activation='tanh'
    )

    # Final output layer: 3 hidden units -> 1 output
    w3 = m.bnn.layer_linear(
        X=w2,
        dist=m.dist.normal(0, 1, name='w3', shape=(D_H2, 1))
    )

    # Squeeze the final output to match Y's shape of (N,)
    logits = w3.squeeze(-1)

    # Likelihood mapping the logits to binary outcomes
    m.dist.binomial(total_count=1, logits=logits, obs=Y, name='Y')

# Run mcmc ------------------------------------------------
m.fit(model, progress_bar=False) # Approximate posterior distributions

# Predictions from the model ------------------------------------------------
import matplotlib.pyplot as plt

# Create a grid to evaluate the model
n_grid = 50
x0 = jnp.linspace(X[:, 0].min() - 0.5, X[:, 0].max() + 0.5, n_grid)
x1 = jnp.linspace(X[:, 1].min() - 0.5, X[:, 1].max() + 0.5, n_grid)
xx0, xx1 = jnp.meshgrid(x0, x1)
X_grid = jnp.c_[xx0.ravel(), xx1.ravel()]

# Swap data on model temporarily to predict on the grid
m.data_on_model = dict(X=X_grid, Y=jnp.zeros(X_grid.shape[0]))
pred = m.sample(samples=500)['Y']
p_mean = jnp.mean(pred, axis=0)

# Plotting the posterior predictive mean
fig, ax = plt.subplots(figsize=(8, 6), constrained_layout=True)
contour = ax.contourf(xx0, xx1, p_mean.reshape(n_grid, n_grid), cmap="coolwarm", alpha=0.6)
scatter = ax.scatter(X[:, 0], X[:, 1], c=Y, cmap="coolwarm", edgecolors='k')
ax.set(title="Posterior Predictive Mean", xlabel="Feature 1", ylabel="Feature 2")
fig.colorbar(contour, ax=ax)
```

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jax.local_device_count 16
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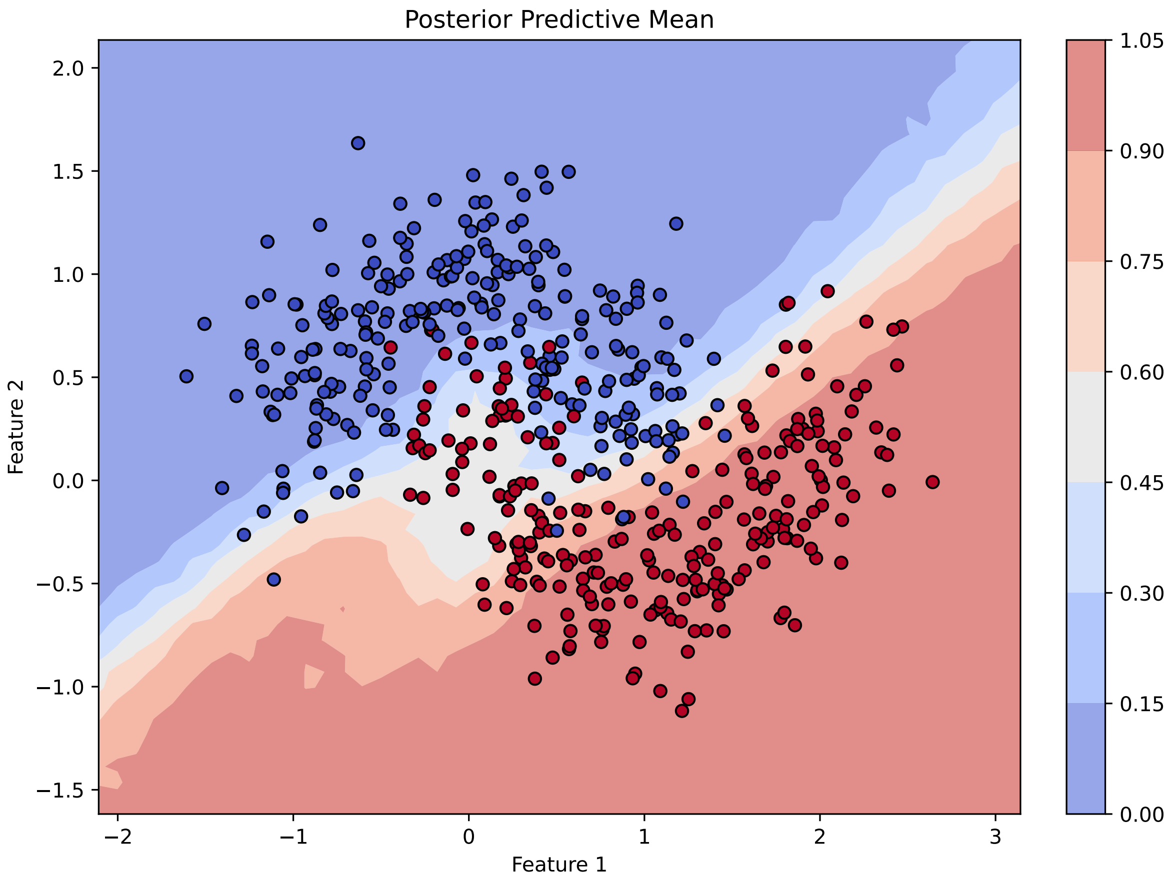
```



Julia

```
using BayesianInference
using PyCall

# Setup device------------------------------------------------
m = importBI(platform="cpu")

# Generate Synthetic Data using Python's scikit-learn ---------
sk_datasets = pyimport("sklearn.datasets")
X = sk_datasets.make_moons(n_samples=500, noise=0.25, random_state=42)[1]
Y = sk_datasets.make_moons(n_samples=500, noise=0.25, random_state=42)[2]

m.data_on_model["X"] = X
m.data_on_model["Y"] = Y

# Define model -----------------------------------------------
@BI function model(X, Y)
    N, D_X = size(X)
    D_H1 = 4
    D_H2 = 3

    # First hidden layer
    w1 = m.bnn.layer_linear(
        X,
        dist=m.dist.normal(0, 1, name="w1", shape=(D_X, D_H1)),
        activation="tanh"
    )

    # Second hidden layer
    w2 = m.bnn.layer_linear(
        w1,
        dist=m.dist.normal(0, 1, name="w2", shape=(D_H1, D_H2)),
        activation="tanh"
    )

    # Final output layer
    w3 = m.bnn.layer_linear(
        w2,
        dist=m.dist.normal(0, 1, name="w3", shape=(D_H2, 1))
    )

    # Extract logits
    logits = w3[:, 1]
```

```
    # Likelihood mapping the logits to binary outcomes
    m.dist.binomial(total_count=1, logits=logits, obs=Y, name="Y")
end

# Run mcmc ------------------------------------------------
m.fit(model, num_samples=500, progress_bar=false)
```

Mathematical Details

Bayesian Formulation

For a binary classification task spanning N observations, we model the probability p_i that the response $Y_i \in \{0, 1\}$ is 1. The input features X are passed through a series of hidden layers mapping to this probability:

1. **First Hidden Layer**:
$$\Omega_{i,1} = \tanh(X_i \Theta_1)$$
Where X_i is a 2-vector of features and Θ_1 is a 2×4 weight matrix parameterized by a Standard Normal prior: $\Theta_1 \sim \text{Normal}(0, 1)$.

2. **Second Hidden Layer**:
$$\Omega_{i,2} = \tanh(\Omega_{i,1} \Theta_2)$$
Where Θ_2 is a 4×3 weight matrix parameterized by a Standard Normal prior: $\Theta_2 \sim \text{Normal}(0, 1)$.

3. **Output Layer (Logits)**:
$$\text{logit}(p_i) = \Omega_{i,2} \Theta_3$$
Where Θ_3 is a 3×1 weight matrix parameterized by a Standard Normal prior: $\Theta_3 \sim \text{Normal}(0, 1)$.

4. **Likelihood**:
$$Y_i \sim \text{Binomial}(1, p_i)$$
Here, the model predicts Y_i using a Binomial distribution with 1 trial (equivalent to a Bernoulli distribution). The logits calculated from the network are implicitly passed through the sigmoid function to represent the expected probability p_i.

Notes

> **Note**
>
> - Using `m.dist.binomial(total_count=1)` alongside logit inputs is equivalent to specifying a standard binary cross-entropy loss with a sigmoid activation function in deep learning frameworks.
> - The tutorial uses Stochastic Variational Inference (SVI) since exact MCMC approaches struggle with high-dimensional highly-correlated posterior spaces typical to neural networks. If you find MCMC chains not mixing during `m.fit()`, consider switching to an SVI backend or modifying priors to enforce further regularization.

Reference(s)

1. PyData Berlin 2025: Introduction to Stochastic Variational Inference with NumPyro